Shared evidence | Idealised exchange comparison

Exchange and ventilation: two linked jobs

Name ____________________________ Date _______________

All values are relative model inputs, not measured human data. Hold the omitted proportionality constant the same.

Case	Relative area	Relative concentration difference	Relative thickness
P	2	4	1
Q	4	4	1
R	2	4	2

Model rule: relative rate index=area×concentration difference÷thickness. It compares directions and ratios; it has no calibrated physiological unit.

Air movement case

During quiet inhalation, diaphragm contraction and rib movement increase thoracic volume. Compare pressure inside with surrounding air.

Gradient case

Oxygen is moving from alveolar air into arriving blood. In this scenario, fresh air replacement is reduced while exchange continues.

W1. Area-only pair and prediction. W2. Thickness-only pair and prediction. W3. Explain the fresh-air effect.

Supported route | Connect the three jobs

Word bank: ventilation; diffusion; blood transport; respiration; gradient; distance.

S1. Draw oxygen and carbon-dioxide arrows across an alveolar surface. Label air and blood.

S2. Explain how diaphragm contraction can make air enter during inhalation.

S3. Compare P/Q and P/R. Name the changed factor and predict the faster/slower model case.

S4. Explain one alveolar adaptation and distinguish ventilation from cellular respiration.

Standard route | Structure explains exchange

Use causal links rather than listing features alone.

M1. Explain the roles of ventilation and blood flow in maintaining gas gradients.

M2. Explain how large area and a thin, moist barrier support gas exchange.

M3. Describe quiet inhalation and exhalation using muscles, volume, pressure and air direction.

M4. Use the table to justify two one-factor comparisons and state why its indices are not clinical readings.

Stretch route | Quantify a model carefully

Optional Biology-specific calculation; Fick-style reasoning is not labelled Higher-only. Use the supplied model rule.

T1. Calculate relative indices for P, Q and R. Compare Q with P and R with P using ratios.

T2. A new case doubles area and doubles thickness, with the same concentration difference. Predict its rate index relative to the original and explain.

T3. A cube of side 1 unit has area 6 and volume 1; a cube of side 2 units has area 24 and volume 8. Calculate both surface-area:volume ratios and explain one implication for exchange.

T4. State two limitations of using this idealised model to explain a real lung.

Exit and reflection | Say how and why

Use scientific process words precisely.

E1. What happens to thoracic pressure during inhalation, and why does air enter?

E2. Why can a thin exchange barrier increase diffusion rate?

E3. Distinguish ventilation or gas diffusion from cellular respiration.

L1. What did your audience ask? Record one stronger mechanism explanation.

Next useful step

A. Explain volume and pressure.

B. Trace separate gas gradients.

C. Link structure with exchange.

D. Limit a model-based claim.